



S Lab Sync

Science Lab Management System — Project Report
Java Swing Desktop Application with MySQL Backend

1. What This Program Does

S Lab Sync is a desktop application built to help a school or college science lab keep track of everything happening in it: the students who use the lab, the teachers who supervise it, the equipment and chemicals on hand, and the experiments currently in progress. Instead of relying on spreadsheets or paper registers, all of this information lives in a proper MySQL database and is managed through a single, consistent desktop interface.

Core screens:

Screen	Purpose
Overview	Live dashboard: total counts of students, teachers, equipment and chemicals, plus a table of every experiment showing who is doing what right now.
Students	Add, update, and delete student records (name, class/grade, roll number, phone).
Teachers	Add, update, and delete teacher records (name, subject, phone, email).
Equipment	Track lab instruments: name, category, quantity, location, condition, and last maintenance date.
Chemicals	Track chemical inventory: name, formula, quantity, unit, storage location, expiry date, and hazard level.
Experiments	Schedule an experiment, assign a supervising teacher, select which students are taking part, and tag it with a color that is reused on the Overview screen.

2. What's Needed to Run It

The application is a plain runnable Java program, so there is no complex installer needed on the development side — but a few pieces have to be in place for it to actually start and connect to a database.

Software / environment

- **Java 17 or newer** (JRE or JDK) — the app is a standard Java Swing program and runs on Windows, macOS, or Linux wherever Java is installed.
- **MySQL Server** running and reachable (normally on `localhost:3306`), with a MySQL account that has permission to create databases and tables.
- **MySQL Connector/J** — the official JDBC driver jar. This is not bundled by default and has to be downloaded once and placed in the app's `lib/` folder as `mysql-connector-j.jar`.

Files that ship with the app

- `SLabSync.jar` — the compiled application.
- `Run_SLabSync.bat` / `Run_SLabSync.sh` — one-click launchers for Windows and macOS/Linux.
- `logo.ico` / `bundled logo.png` — the app icon, used for the window, the taskbar, and an optional desktop shortcut.
- `lib/mysql-connector-j.jar` — the JDBC driver mentioned above (added by whoever sets the app up).
- `source/` — the full Java source code, organized into `db`, `model`, `dao`, and `ui` packages, in case anyone wants to customize or rebuild it.

What happens automatically on first run

- The app asks once for MySQL login details and remembers them for that computer afterward (saved to `~/.labmanager/db.properties`).
- It creates the `lab_management` database and all six required tables itself — no manual SQL setup required.
- On Windows, it offers to add a desktop shortcut with the app's icon, so future launches are a double-click away.

3. How It's Built (Architecture)

The codebase follows a clean, layered structure rather than mixing everything into one giant file, which makes it easier to read, maintain, and extend.

Layer	Responsibility
model/	Plain data classes representing a Student, Teacher, Equipment item, Chemical, and Experiment.
dao/	Database access logic — one class per table, responsible for reading and writing records.
db/	Connection handling: stored credentials, the schema initializer that creates tables on first run, and the live database connection holder.
ui/	All Swing screens: the DB setup dialog, login screen, main dashboard, and one panel per data screen (Students, Teachers, Equipment, Chemicals, Experiments, Overview).
UITheme.java	A single shared source of truth for colors, fonts, and reusable custom-drawn components (cards, buttons, sidebar, table styling), so the whole app has one consistent look instead of each screen improvising its own styling.

4. What's Good About This Program

Polished, custom interface

- Rather than looking like a default Java Swing app, it uses a custom-drawn glass/gradient design with its own color palette, rounded cards, and a frameless window with custom minimize/maximize/close controls.
- A working **light/dark mode toggle** is built in and applied consistently across every screen, not just the main window.
- Experiments can be color-tagged, and that color is reused on the Overview screen so related sessions are easy to spot at a glance.

Thoughtful first-run experience

- New users are guided through MySQL setup with a dedicated dialog instead of being expected to edit a config file by hand.
- The database and every table are created automatically — there's no separate SQL script to run.
- The optional desktop shortcut prompt (with the app's own icon) makes the app feel like a properly installed program rather than "just a jar file".

Sensible engineering choices

- Clear separation between data (`model`), persistence (`dao/db`), and interface (`ui`) makes the codebase easy to navigate and safe to modify.
- Backward-compatible schema handling — if an older database is detected, the app adds the newer `color_hex` column automatically instead of breaking or discarding existing data.
- Ships with both a runnable jar for quick use and instructions for turning it into a native installer (`.exe`, `.dmg`, `.deb`) via `jpackage` for a more polished distribution.
- Full source is included and organized cleanly enough that a developer could change the master password, add new fields, or extend a panel without having to reverse-engineer the whole project.

Practical, everyday usefulness

Beyond the technical implementation, the feature set maps directly onto a real lab's day-to-day needs: knowing what equipment and chemicals are on hand and where, who is supervising which experiment, and which students are involved in it right now. The Overview screen in particular turns that into an at-a-glance snapshot rather than something a lab manager has to piece together from separate records.

This report reflects the current state of the S Lab Sync project, based on its README and Java source code.